L&T

PHYSICS PROJECT REPORT

# SIMULATION AND ANALYSIS OF A FULL-WAVE RECTIFIER USING PN-JUNCTION DIODES

Behaviour, rectification efficiency, ripple factor, and comparison with a half-wave rectifier

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Submitted by

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## 1. Aim

To simulate and study a full-wave rectifier using PN-junction diodes, explain its operation during both half-cycles of an AC input, and compare its rectification efficiency and ripple factor with those of a half-wave rectifier.

## 2. Objectives

- Understand how PN-junction diodes conduct when forward-biased and block current when reverse-biased.
- Observe how a full-wave rectifier converts both halves of an AC waveform into a unidirectional output.
- Calculate ideal rectification efficiency and ripple factor.
- Compare the output quality and practical performance of full-wave and half-wave rectifiers.

## 3. Components / Simulation Setup

Use a low-voltage AC source, PN-junction diodes, a load resistor ($R_L$), and a ground/reference node. A centre-tapped transformer with two diodes can be used for a centre-tapped full-wave circuit; alternatively, four diodes can be connected as a bridge rectifier. In a circuit simulator such as LTspice, Multisim, or Proteus, connect a transient-analysis probe across the load and plot the input and output voltages.

### Suggested simulation settings

| Parameter | Suggested value |
|---|---|
| Input source | 12 V peak, 50 Hz sine wave (low-voltage example) |
| Load resistor | 1 kΩ |
| Diodes | Silicon PN-junction diodes (e.g., 1N400x model) |
| Transient analysis | 0 to 100 ms; sufficiently small time step |

Note: These are example simulation values, not measured results. Actual output depends on circuit topology, diode models, transformer ratio, source impedance, and load.

## 4. Theory and Working

A PN-junction diode allows substantial current in the forward direction once its forward voltage is sufficient, and strongly opposes current in reverse bias (until breakdown). This one-way conduction is used to make the load current flow in the same direction during both halves of an AC cycle.

### Centre-tapped full-wave rectifier

During the positive half-cycle, one end of the centre-tapped secondary becomes positive relative to the centre tap. The corresponding diode conducts, while the other diode is reverse-biased. During the negative half-cycle, the secondary polarity reverses; the other diode conducts and the first blocks. The load-current direction remains the same in both cases.

### Bridge full-wave rectifier

A bridge uses four diodes. Two diodes conduct during one half-cycle and the other two conduct during the next. The load current retains the same direction. A bridge does not require a centre-tapped transformer, but current passes through two forward-biased diodes on each conducting path, increasing the diode voltage drop.

## 5. Expected Waveforms and Behaviour

The input is a sine wave with positive and negative half-cycles. For an ideal full-wave rectifier, the output is approximately the absolute value of the input, so both half-cycles appear above the zero-voltage axis. The ripple pulses occur twice per input cycle; therefore, a 50 Hz supply gives a 100 Hz ripple frequency.

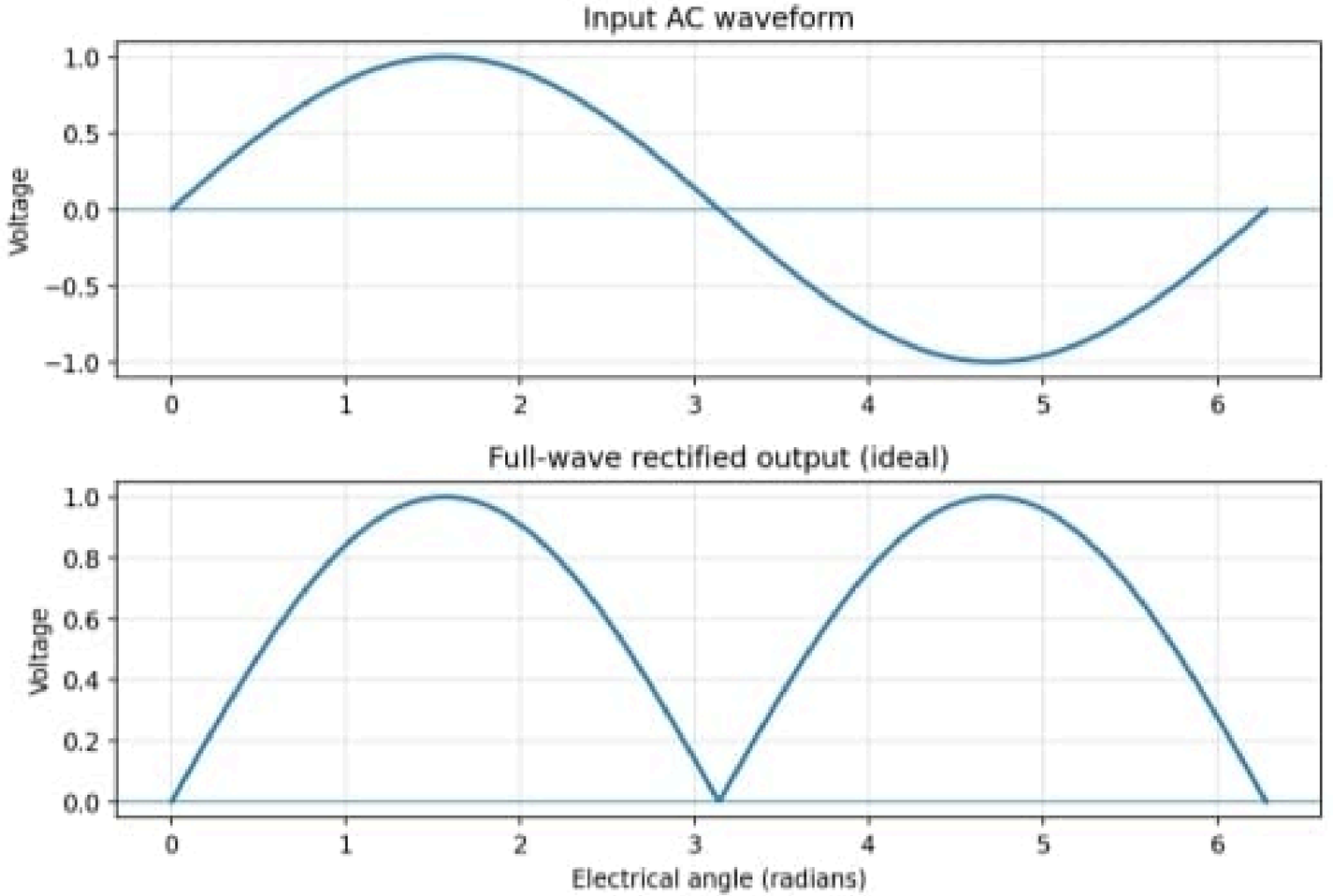



Figure 1. Ideal input and full-wave rectified output waveforms.

Rectification efficiency (η) is the ratio of DC power delivered to the load to the AC input power supplied to the rectifier, expressed as a percentage:

$$\eta = P_{DC} / P_{AC} \times 100\%$$

For an ideal rectifier with a resistive load, the standard maximum efficiencies are obtained from the average (DC) and RMS output values. For a half-wave rectifier, $V_{DC} = V_m/\pi$ and $V_{RMS} = V_m/2$. For a full-wave rectifier, $V_{DC} = 2V_m/\pi$ and $V_{RMS} = V_m/\sqrt{2}$. Using $\eta = (V_{DC}/V_{RMS})^2$ for the same resistive load gives:

$$\text{Half-wave: } \eta_{max} = 40.6\%$$
$$\text{Full-wave: } \eta_{max} = 81.2\%$$

Thus, the ideal full-wave rectifier is about twice as efficient as the half-wave rectifier. In real circuits, diode forward drops, transformer losses, winding resistance, and source impedance reduce the efficiency. In a bridge rectifier, two diode drops occur in each conducting path; in a centre-tapped circuit, one diode drop occurs but transformer utilisation differs.

## 7. Ripple Factor

Ripple factor (r) measures the amount of AC variation remaining in the rectified output relative to its DC component. A smaller ripple factor means a smoother DC output. Without a filter capacitor, the standard ideal values are:

$$r = \sqrt{[(V_{RMS} / V_{DC})^2 - 1]}$$
$$\text{Half-wave: } r \approx 1.21$$
$$\text{Full-wave: } r \approx 0.482$$

The full-wave output has a lower ripple factor because both input half-cycles contribute to the output, producing more frequent pulses and a larger average value relative to the ripple. A capacitor or other filter can reduce ripple further; the actual filtered ripple depends on load current, capacitance, frequency, and circuit resistance.

## 8. Comparison: Half-Wave vs Full-Wave

| Feature | Half-wave rectifier | Full-wave rectifier |
|---|---|---|
| Diodes used | 1 | 2 (centre-tapped) or 4 (bridge) |
| Input half-cycles used | One | Both |
| Ripple frequency | f | 2f |
| Maximum ideal efficiency | 40.6% | 81.2% |
| Ripple factor (no filter) | 1.21 | 0.482 |
| Output smoothness | More ripple | Less ripple |
| Circuit complexity | Simple | More components |

## 9. Simulation Procedure

1. Open LTspice, Multisim, Proteus, or another circuit simulator.
2. Build either a centre-tapped full-wave rectifier or a bridge rectifier using PN-junction diode models.
3. Connect the load resistor across the rectifier output and connect the AC source to the transformer secondary or bridge input.

cycles.
5. Run the simulation and plot the input voltage and load voltage on the same time axis.
6. Verify that output pulses appear during both half-cycles and that the pulse frequency is twice the input frequency.
7. Measure the average and RMS output values using the simulator's measurement tools. Calculate $\eta$ and r, then compare with the ideal values above. Repeat with a half-wave circuit using the same source and load.

## 10. Observations and Discussion

Expected observations: (i) the full-wave output remains the same polarity; (ii) there are two output pulses per input cycle; (iii) the full-wave average output is higher than that of a half-wave rectifier for the same peak input in the ideal case; and (iv) full-wave rectification has higher ideal efficiency and lower unfiltered ripple factor. In a real simulation, the output peaks will be lower than the ideal waveform because of diode forward voltage and other circuit losses.

For a fair comparison, use the same AC frequency, comparable peak voltage and load resistance, and clearly state whether the model is a centre-tapped or bridge circuit. Record the simulator's measured values rather than presenting theoretical predictions as experimental measurements.

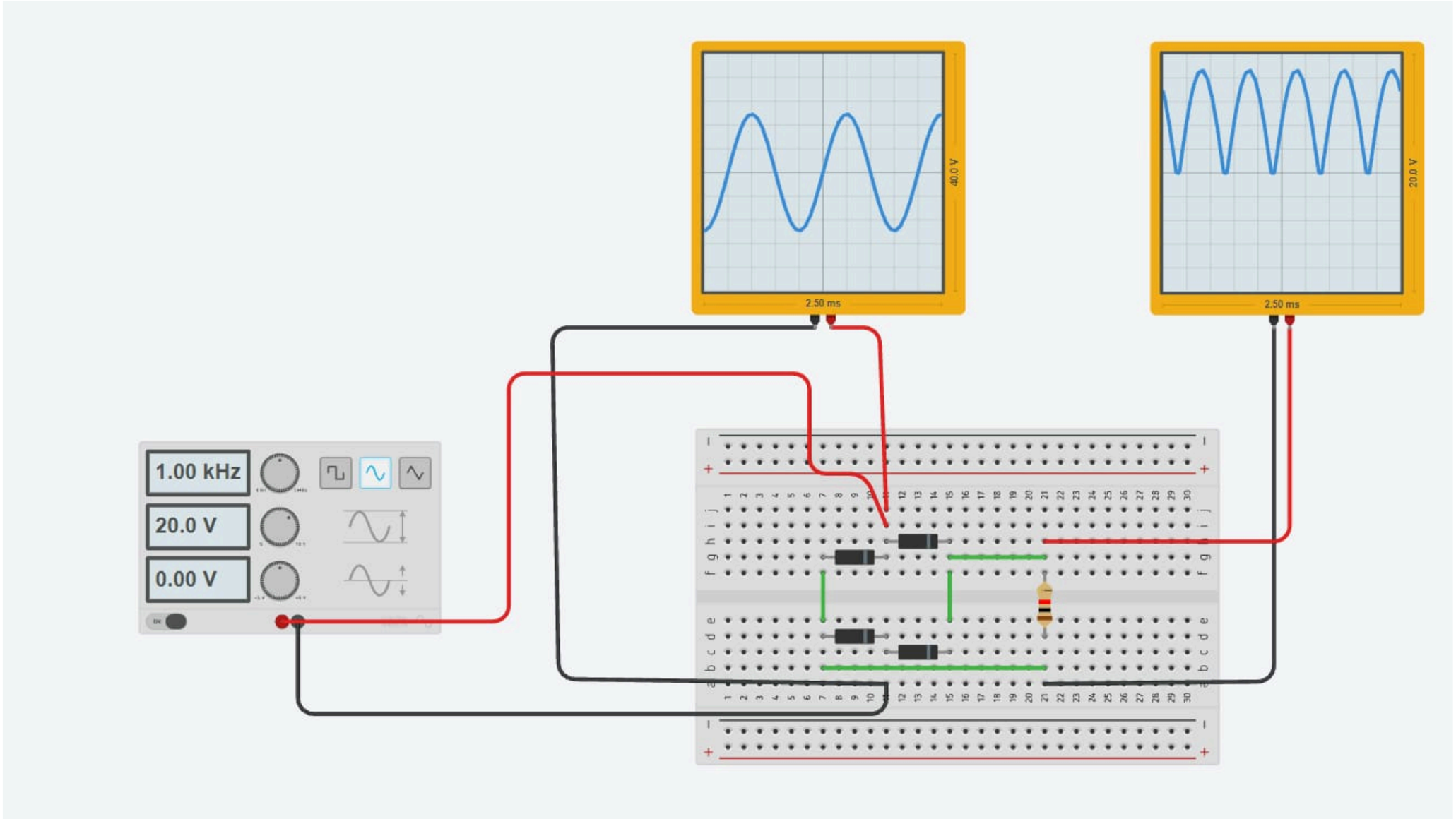
## 11. Conclusion

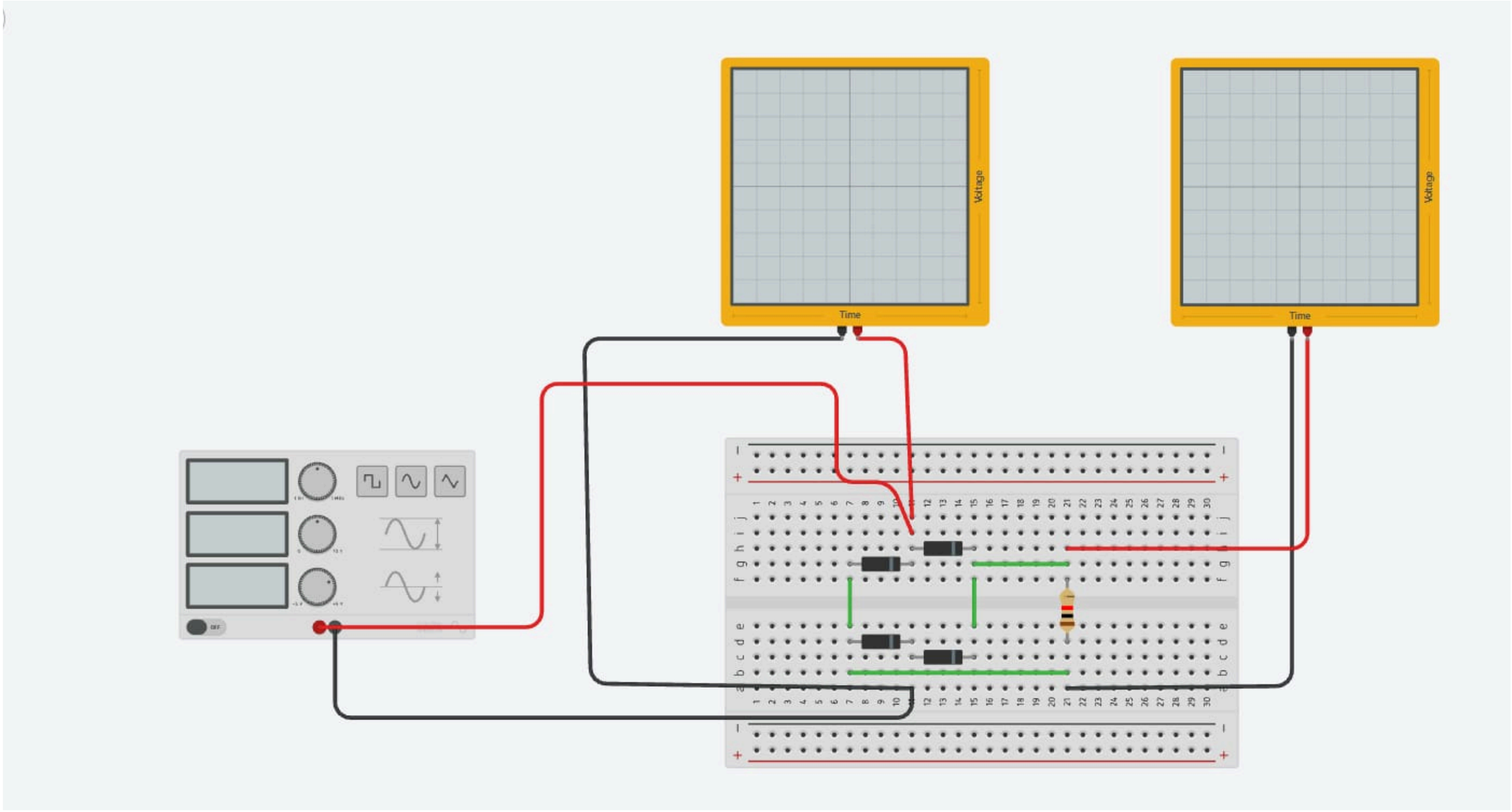
A PN-junction full-wave rectifier converts both halves of an AC input into a unidirectional, pulsating DC output. Compared with a half-wave rectifier, it uses the input more effectively, has a ripple frequency twice as high, offers a theoretical maximum rectification efficiency of about 81.2% rather than 40.6%, and has a lower unfiltered ripple factor of about 0.482 rather than 1.21. These advantages make full-wave rectifiers more suitable for DC power supplies, although filtering and regulation are usually needed for a smooth, stable DC output.

## 12. References


- B. L. Theraja and A. K. Theraja, *A Textbook of Electrical Technology*, Vol. I, S. Chand Publishing.
- Robert L. Boylestad, *Introductory Circuit Analysis*, Pearson.
- Diode and transient-analysis documentation for the circuit simulator used (for example, LTspice or NI Multisim).


End of Report


40.0 V
2.50 ms
20.0 V
2.50 ms
1.00 kHz
20.0 V
0.00 V


Voltage
Time
Voltage
Time

https://www.tinkercad.com/things/5rUe0nstOw4-full-wave-rectifier

# THANK YOU

